

BTP: EMI Effects in Voltage-to-Time Converters

Analysis, Cadence exploration, and differential EMI-cancellation architecture

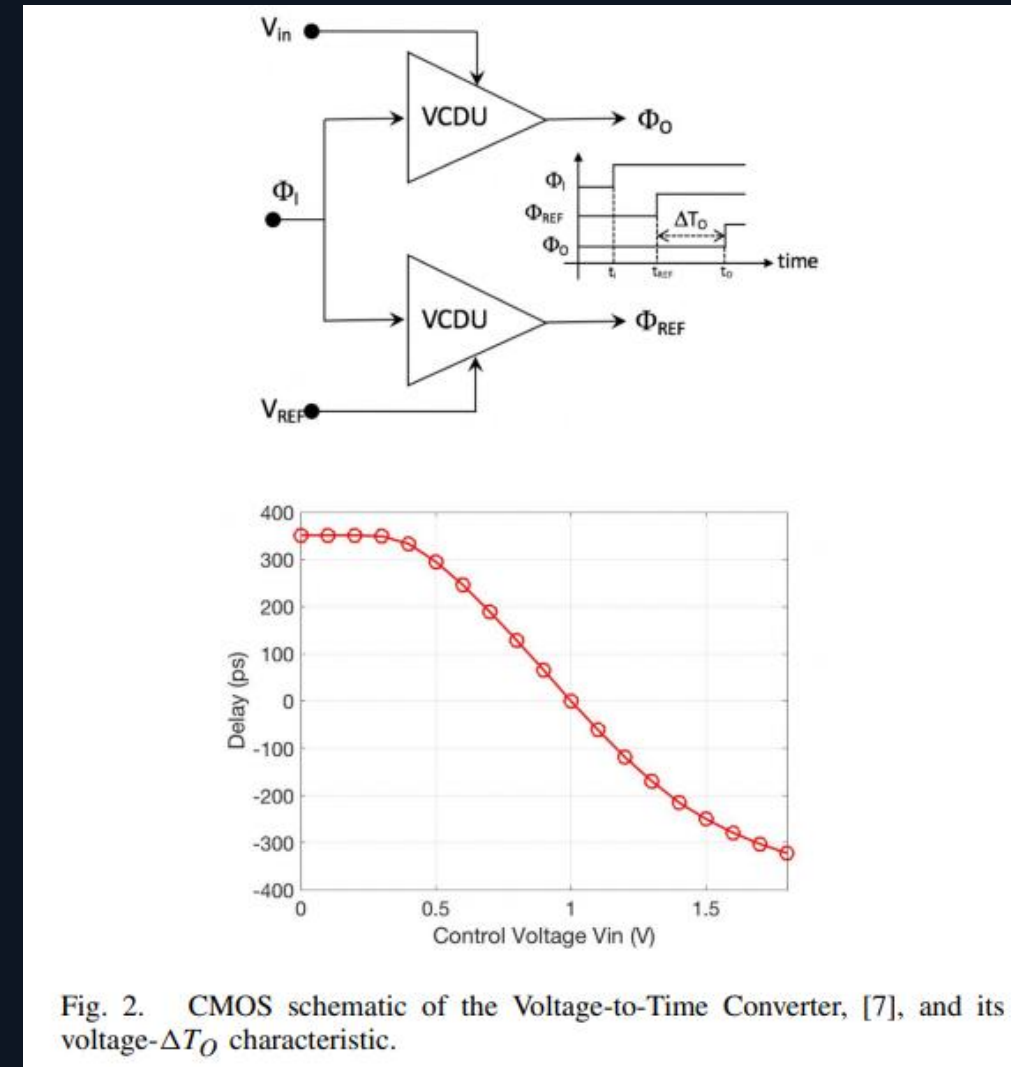
G. Jaidev | 2023EEB1200

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Department of Electrical Engineering, IIT Ropar

Project Story: From VTC to EMI-Robust Architecture

Before mid-semester: common VCDU/VTC foundation, PVT study, Monte Carlo and EMI base-paper analysis.
After mid-semester: BTP diverged toward EMI reduction using differential delay architectures.
Main research question: how can common-mode EMI be reduced after voltage is converted to time?
Final direction: compressed small-signal differential control around 0.9 V.



Baseline VTC from literature: two VCDU paths generate a time difference.

Pre-Midsemester Foundation

VCDU/VTC operation, PVT variation, and baseline EMI susceptibility

Baseline VCDU/VTC Principle

A VCDU delays an input edge according to a control voltage.

Two VCDUs form a VTC: one input-controlled path and one reference path.

Output is measured as a time difference, not an absolute voltage.

This structure is useful for TMSP but still sensitive to timing disturbances.

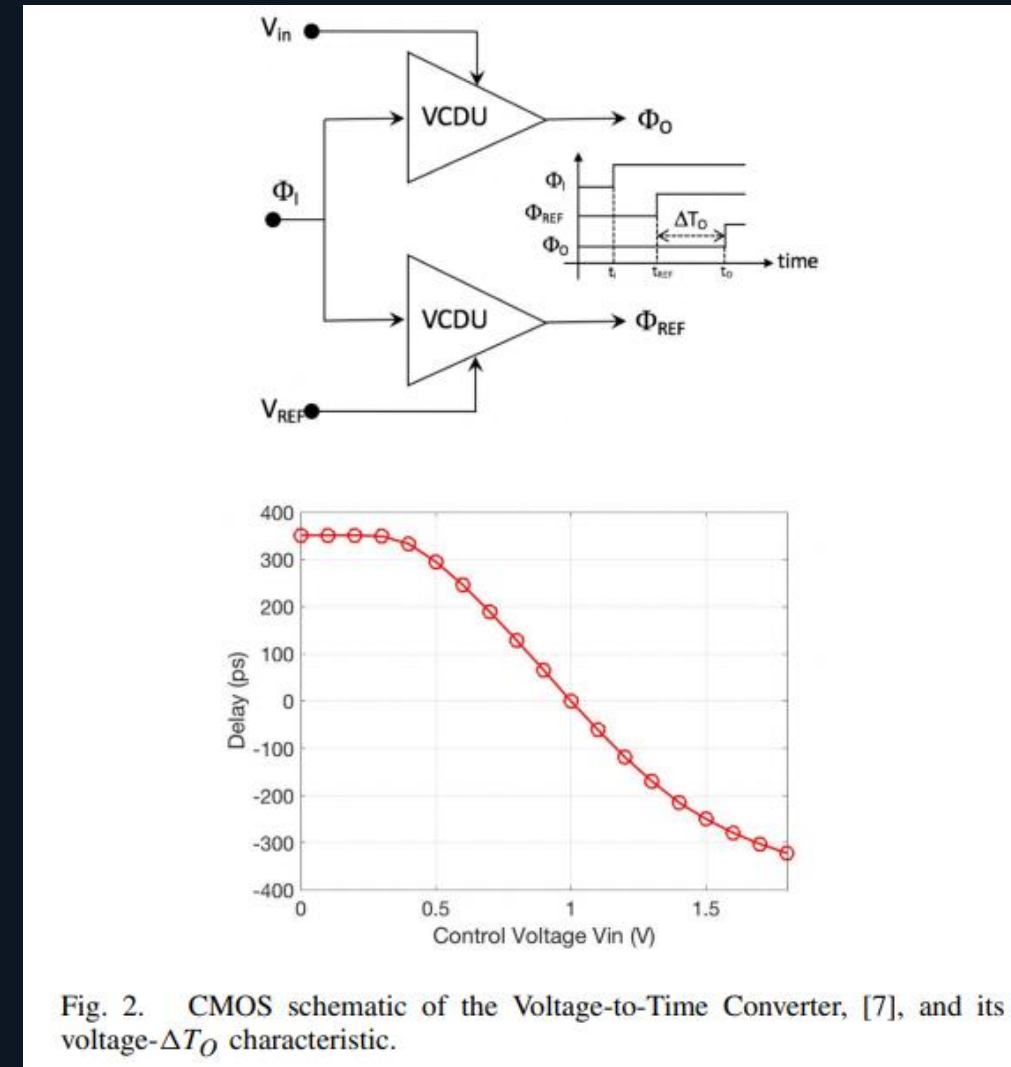


Fig. 2. CMOS schematic of the Voltage-to-Time Converter, [7], and its voltage- ΔT_O characteristic.

Pre-midsem baseline VTC architecture.

Delay Model Used Throughout BTP

$$I_{\text{dis}} = -C_L \, dV_C/dt$$

$$t_d = C_L \, \Delta V / I_{\text{ctrl}}$$

$$\Delta I \approx g_m \, \Delta V_{\text{ctrl}}$$

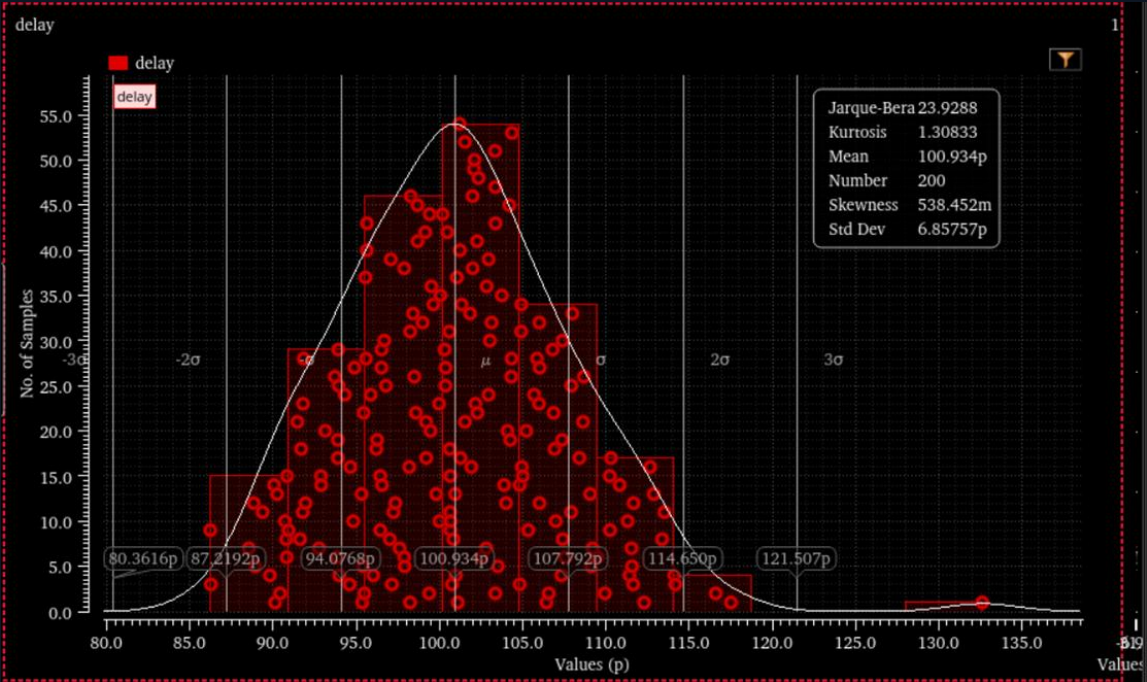
$$\Delta t_d \approx -(C_L \, \Delta V / I_0^2) \, \Delta I$$

Delay is the capacitor discharge time until the inverter threshold is crossed.

Any perturbation in current or threshold appears as timing jitter.

PVT and Monte Carlo: Single VCDU Sensitivity

Single VCDU delay varies strongly with process, voltage, temperature and mismatch; this was established in the mid-semester work.



Monte Carlo distribution / delay spread

Corners	☒ Nominal	☒ TT	☒ FF	☒ SS
Temperature		27	-40	100
Design Variables				
v		0.4:0.1:1.4	0.4:0.1:1.4	0.4:0.1:1.4
vdd		1.8	1.98	1.62
Click to add				
Parameters				
Click to add				
Model Files				
gpdk.scs	☒	NN	☒ FF	☒ SS
Click to add				
Model Group(s)		<modelgroup>	<modelgroup>	<modelgroup>
Click to add				

Cadence summary table

Why Dual VCDU Helps With PVT

Single VCDU: absolute delay includes PVT shift.
Dual VCDU/VTC: output is a difference of two similar paths.
Common PVT delay shifts cancel approximately in Delta T.
This improves timing stability, but does not cancel unequal EMI coupling.

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DUAL VCDU (VTC) PVT AND MONTE CARLO ANALYSIS

- Single VCDU: Delay = f(Vin, PVT)
- Dual VCDU: $\Delta T = f(Vin) - f(Vref)$
- Common PVT variations cancel out

ΔT vs Vin shows reduced variation
Curves for TT, SS, FF nearly overlap
Improved stability

Reduced standard deviation
Narrower distribution
Better immunity to mismatch

Parameter	Single VCDU	Dual VCDU
PVT Sensitivity	High	Low
Delay Variation	large	small
Robustness	Poor	Good

- Single VCDU shows strong PVT dependence
- Dual VCDU significantly reduces variation
- Monte Carlo confirms improved robustness
- Suitable for precise timing applications

Corners	Nominal	TT	FF	SS
Temperature		27	-40	100
Design Variables				
v		0.4:0.1:1.4	0.4:0.1:1.4	0.4:0.1:1.4
vdd		1.8	1.98	1.62
Click to add				
Parameters				
Click to add				
Model Files				
gpdic.scs		NN	FF	SS
Click to add				
Model Group(s)	<modelgroup>	<modelgroup>	<modelgroup>	<modelgroup>

Baseline EMI Injection: Core Observation

EMI was modelled as a sinusoidal disturbance. Amplitude, phase and frequency all affect delay.

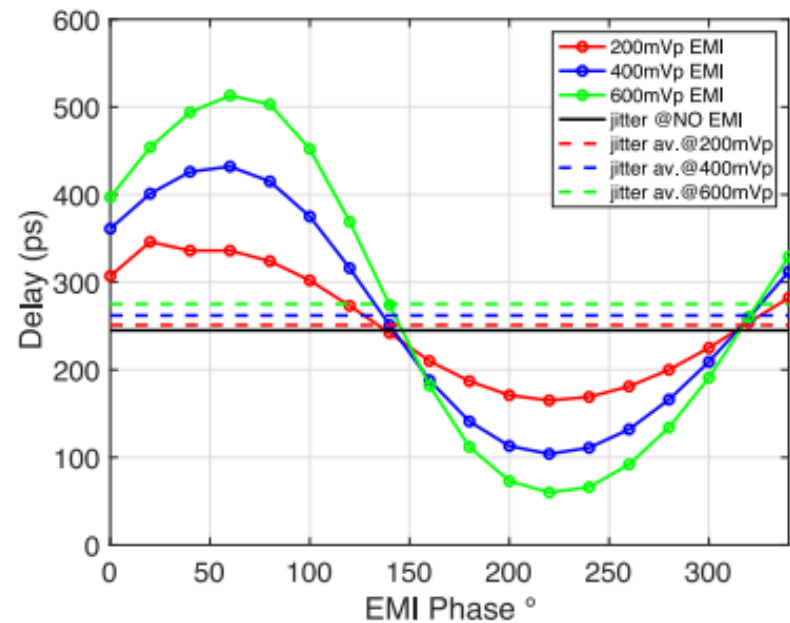


Fig. 4. ΔT_o @ $V_{in}=0.6V$, @ $V_{REF}=1V+EMI$, @EMI freq=500MHz.

EMI injected at V_in

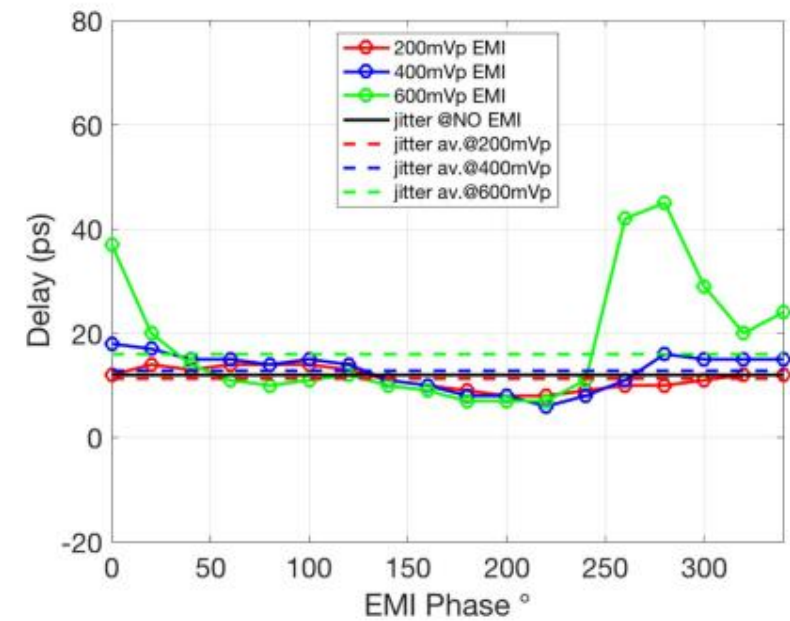


Fig. 13. ΔT_o @ $V_{in}=0.6V+EMI$; $V_{REF}=1.2V+EMI$, $V_{dd}=1.8V+EMI$, $V_{ss}=0V+EMI$, @EMI freq=500MHz.

All-pin EMI scenario

Why EMI Phase Matters

$$v_{emi}(t) = A_{emi} \sin(2\pi f_{emi} t + \phi)$$

$$v_{emi}(t_s) = A_{emi} \sin(2\pi f_{emi} t_s + \phi)$$

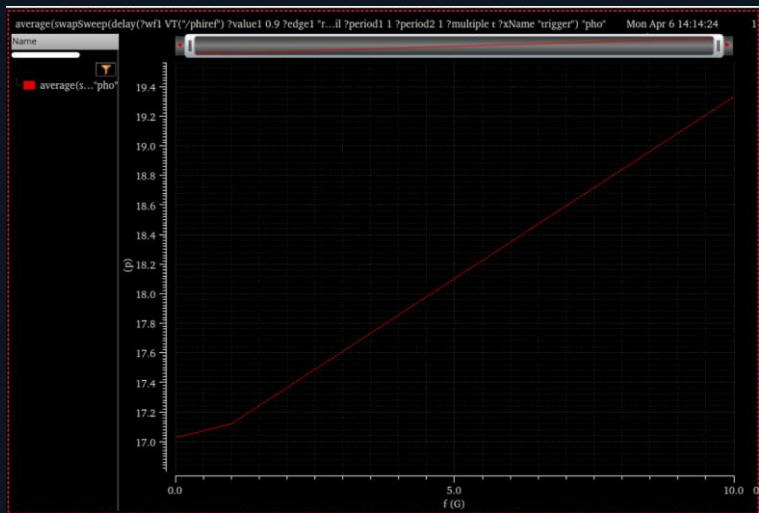
Delta t_d proportional to $v_{emi}(t_s)$

Worst-case: $\max_{\phi} |\Delta t_d(\phi)|$

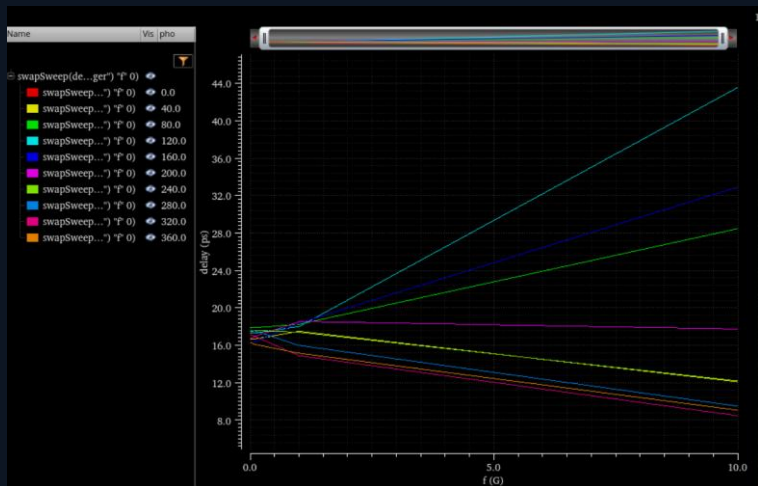
A VTC makes a timing decision at one instant. The EMI value at that instant determines edge advance/delay.

Baseline VTC Frequency/Phase Sweep

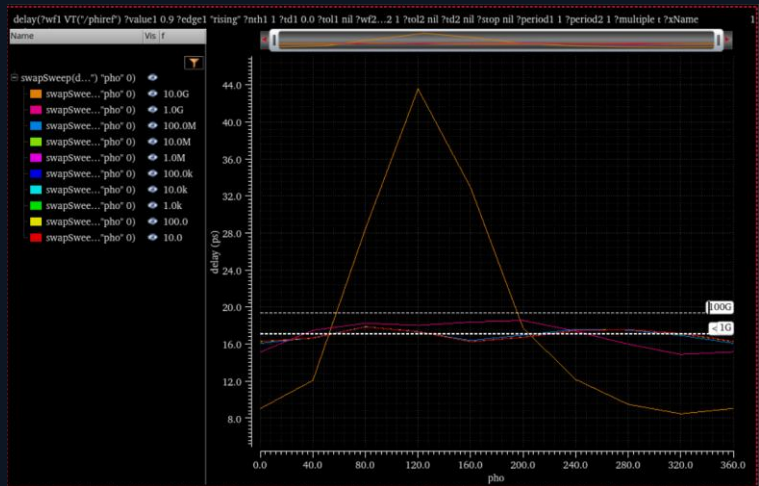
These results motivated the BTP: average delay is not enough; worst-case phase at GHz frequency can be much larger.



Average delay increases with frequency



Delay depends on phase and frequency



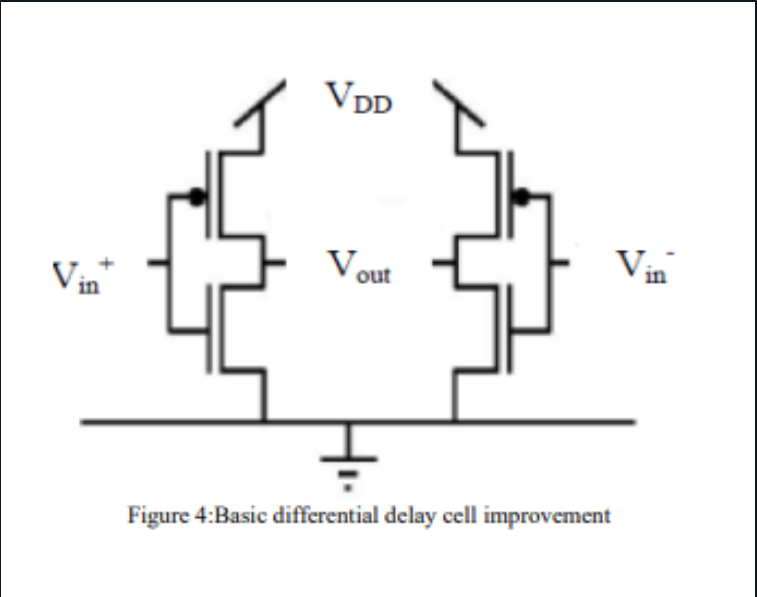
Worst-case phase peak at high frequency

Post-Midsemester Circuit Exploration

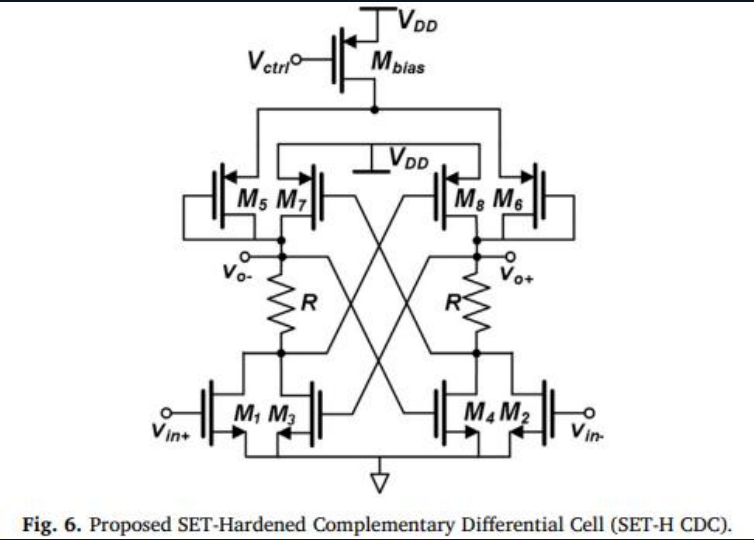
Differential delay cells tried after the scope diverged from DEP

Differential Delay-Cell Literature Survey

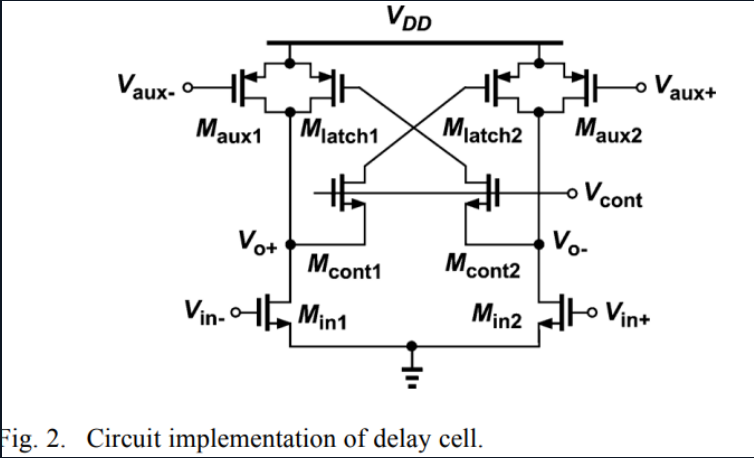
Three topologies were reviewed and implemented/analyzed as candidates before choosing the final direction.



Basic differential delay cell



SET-hardened complementary differential cell



Input-conditioned / dual-path delay cell

Why Basic Differential Delay Was the Starting Point

Differential signalling can reject common-mode noise in principle.
If both branches receive the same EMI, subtraction cancels it.
But the delay cell must still give clean, monotonic voltage-to-time conversion.
This became the selection criterion for later attempts.

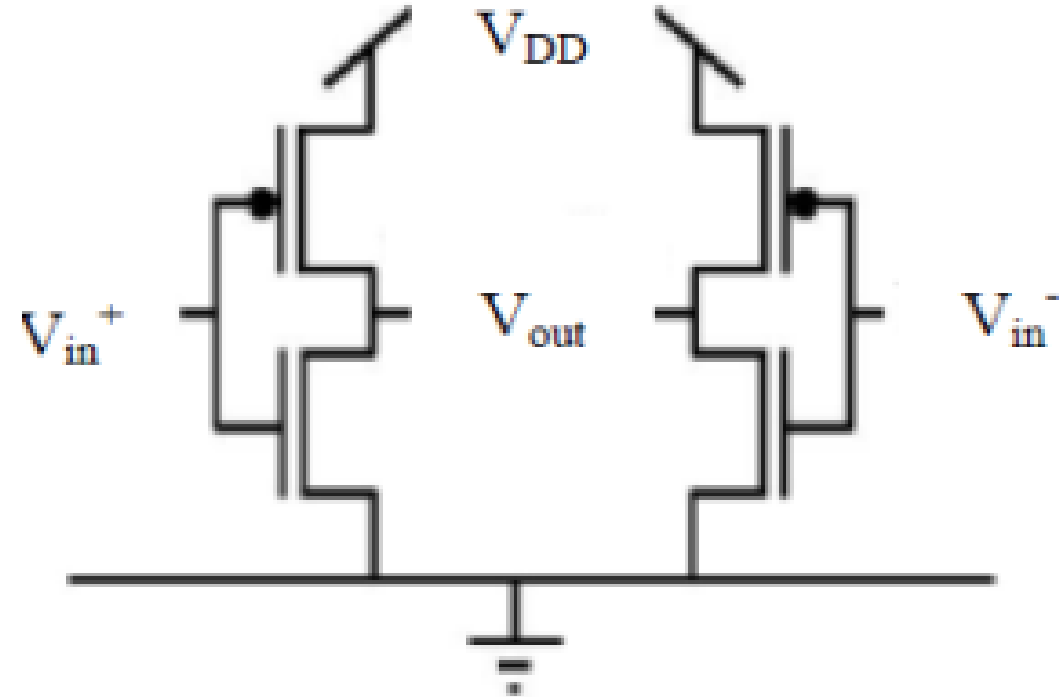
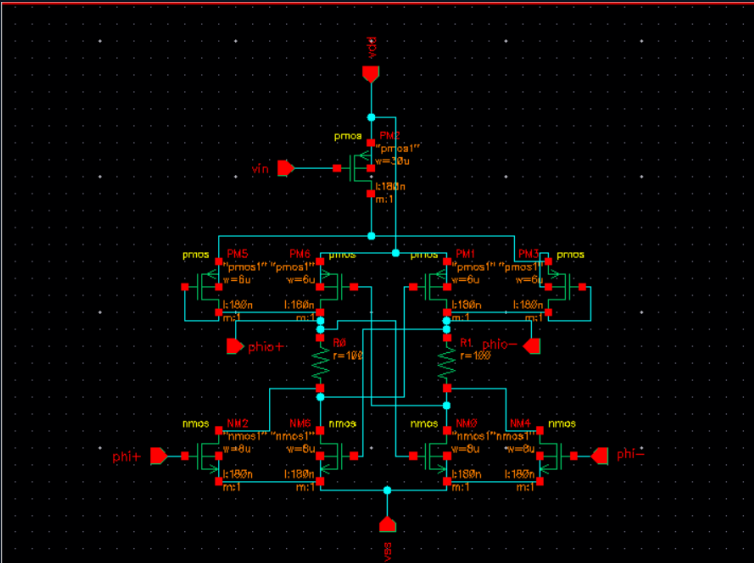


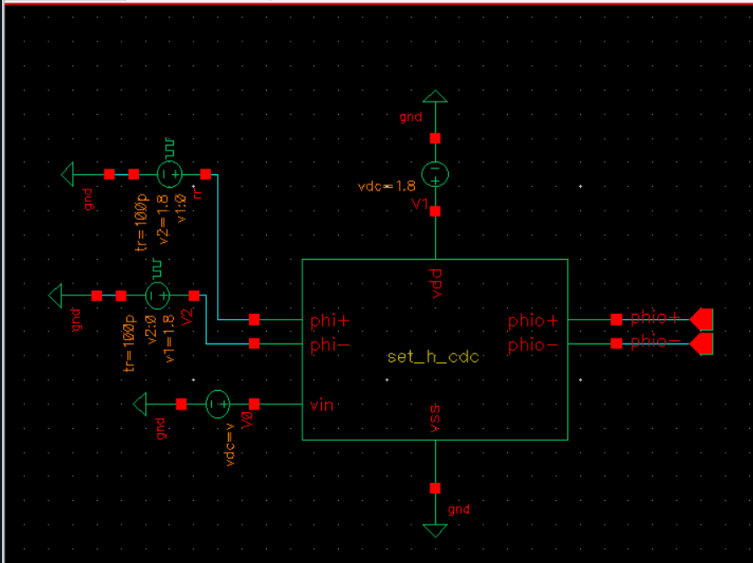
Figure 4: Basic differential delay cell improvement

SET-Hardened CDC Implementation

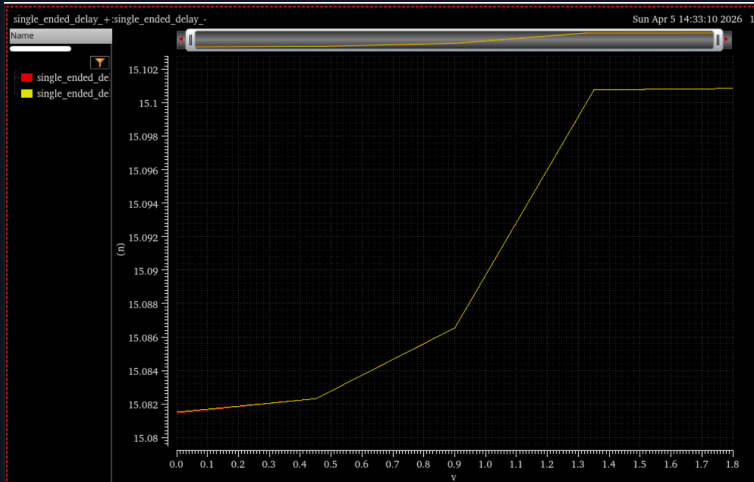
This cell was studied because its cross-coupled feedback gives strong switching and robustness.



Internal Cadence schematic



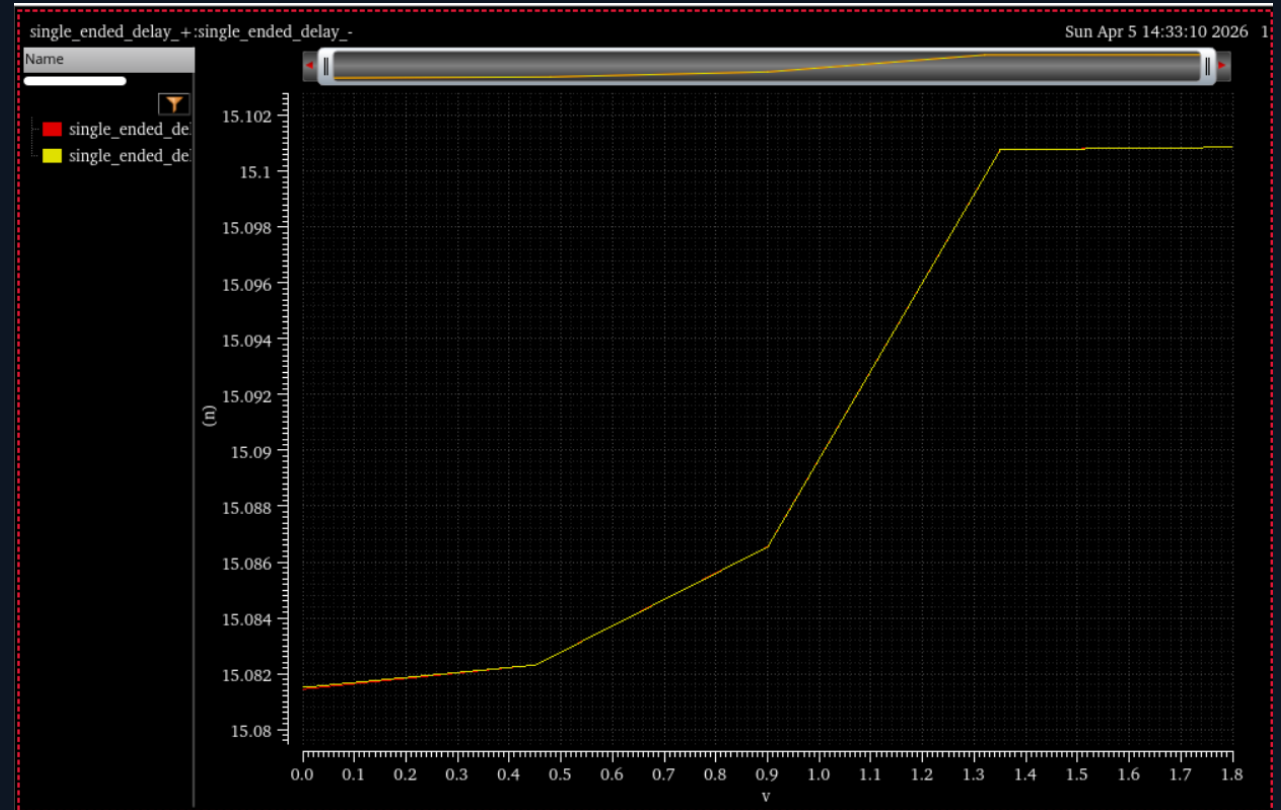
Testbench



Measured delay response

SET-H CDC: Insight and Limitation

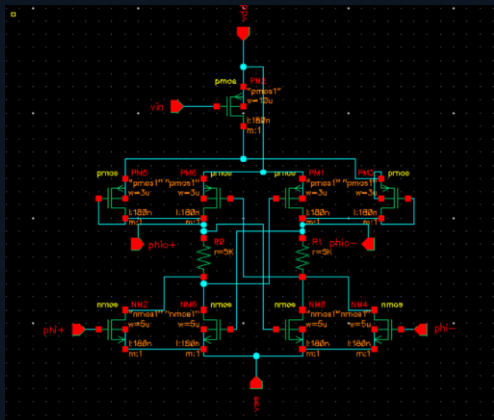
Good for robust/latch-like switching, not automatically good for VTC conversion.
Delay tuning was weak/nonlinear over the input sweep.
Cross-coupled feedback caused abrupt transition behaviour.
Decision: used as design-learning block, not final architecture.



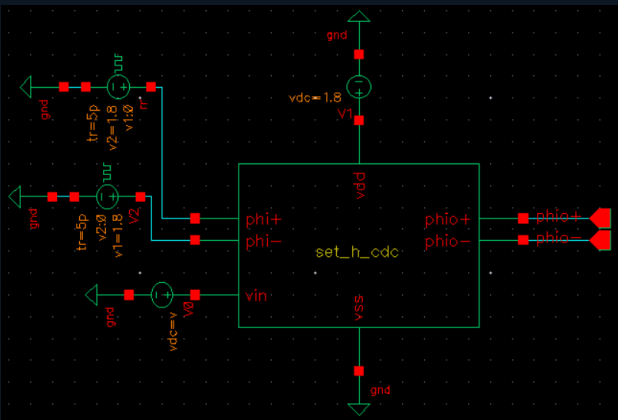
Delay response was not smooth enough for final VTC.

Input-Conditioned Delay Cell Implementation

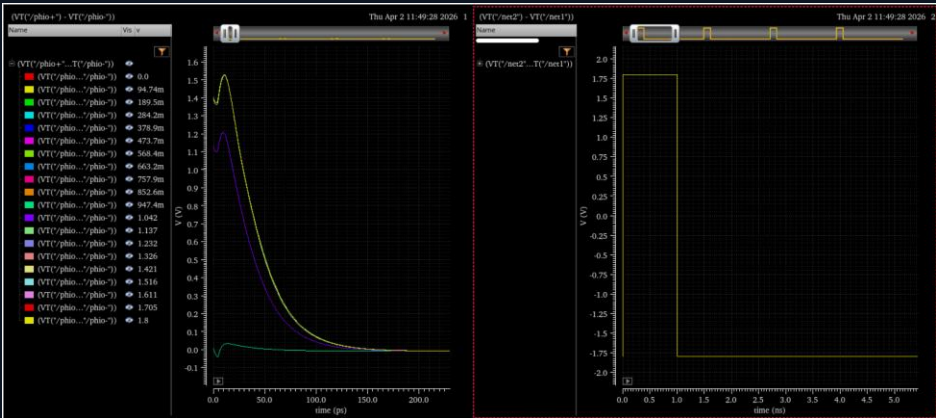
This cell was a separate and important attempt; it was not the same as the SET-H CDC.



Internal schematic



Testbench



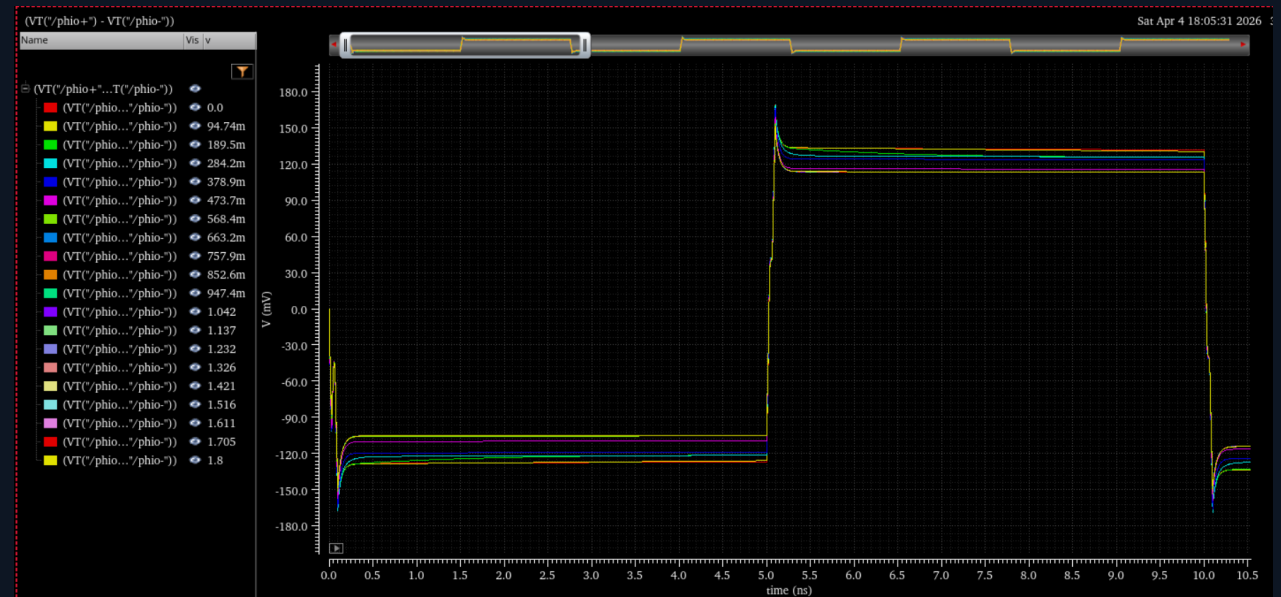
Differential transient



Delay response

Input-Conditioned Cell: Why Rejected

Auxiliary control paths made biasing difficult.
Transient output had overshoot/settling and limited clean swing.
Delay extraction became threshold-dependent.
Decision: not suitable as the main VTC delay core.



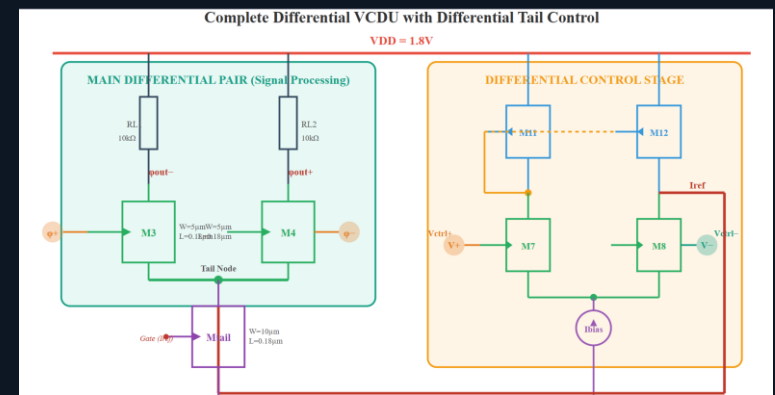
Irregular delay response under sweep.

Custom Differential Attempts

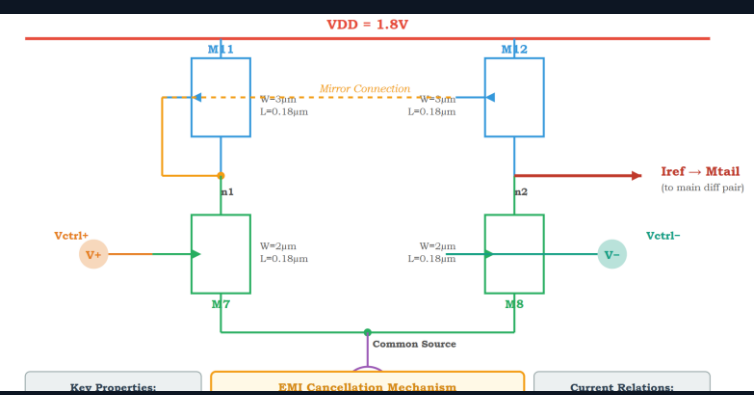
What was built in Cadence and why some versions were rejected

Direct Differential-Pair Controlled Delay Cell

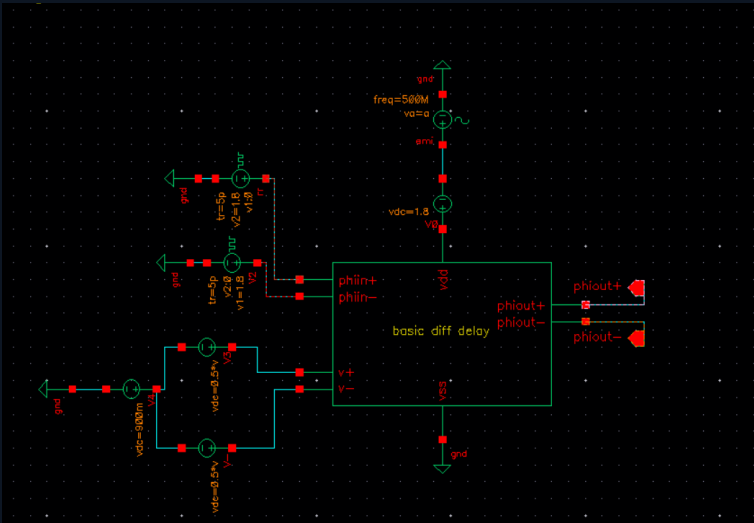
First custom attempt: use a differential pair output to control the differential delay current.



Conceptual differential delay architecture



Expanded schematic



Cadence testbench

Core Problem: Differential Input Became Single-Ended Control

The front-end was differential, but later the control was effectively single-ended.

Common-mode/supply disturbance re-entered the control current.

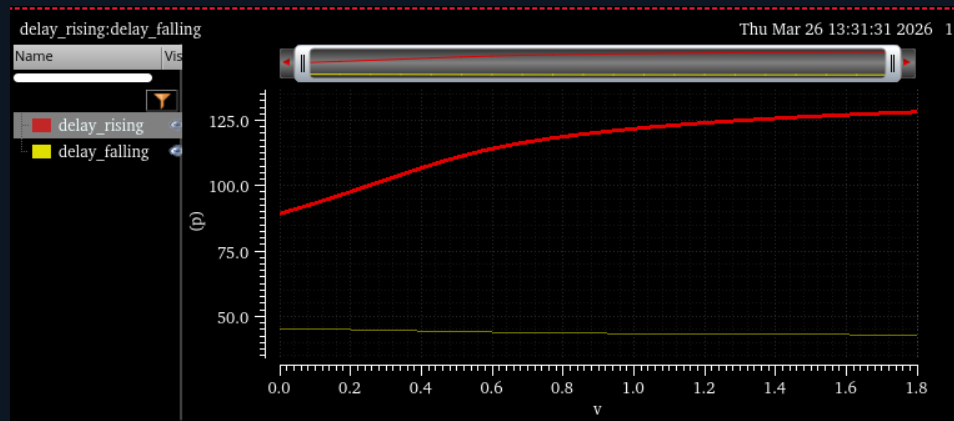
Rising and falling delays became unbalanced.

This explained why differential structure alone was not enough.

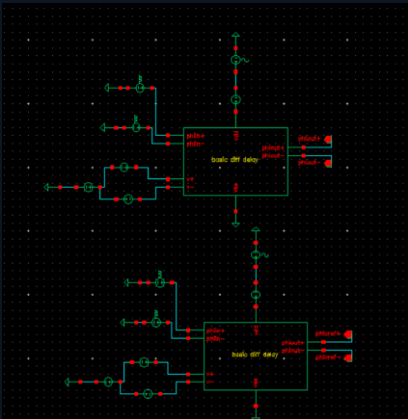
Rising/falling delay mismatch.

Direct Attempt: Waveforms and Delay Issues

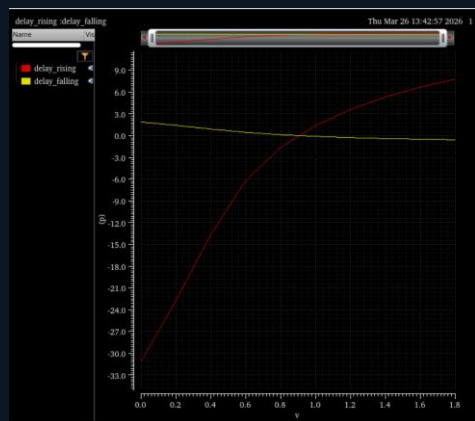
These figures are kept because they explain the actual design challenge, not as final success claims.



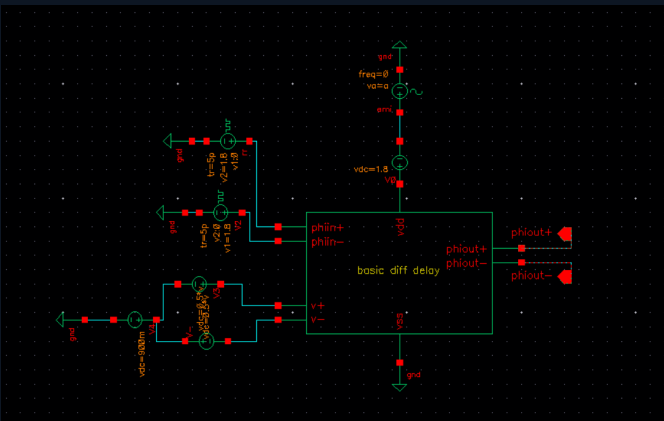
Differential transient output



Two-path VTC testbench



Delay curve



Edge mismatch

EMI Analysis of Intermediate Architecture

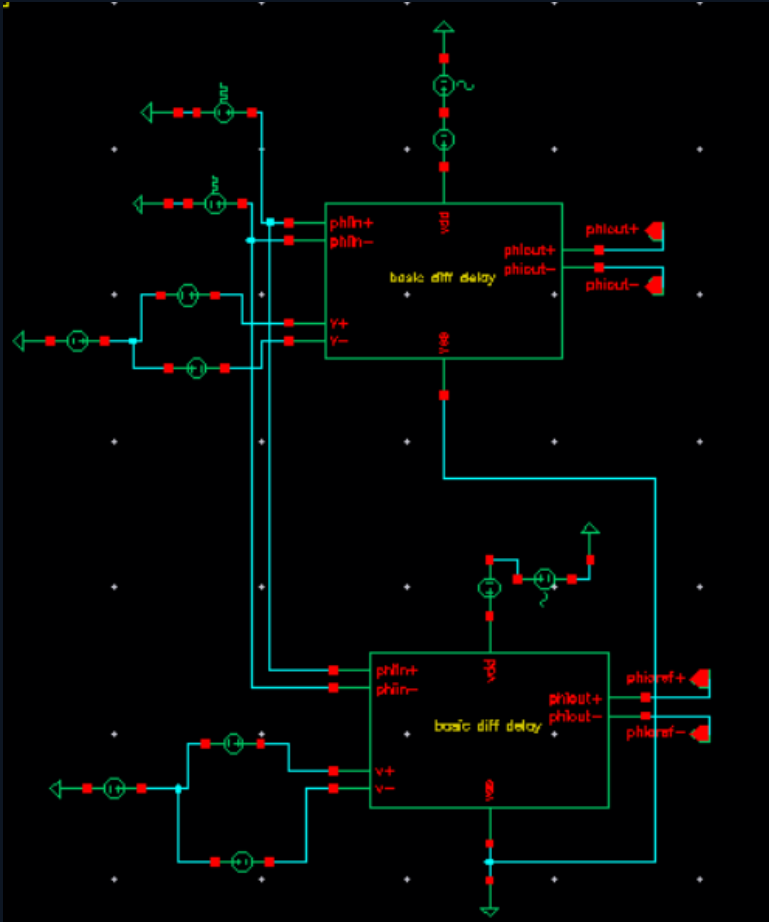
Systematic sweeps across amplitude, phase, control voltage and injection ports

EMI Sweep Setup

Amplitude, phase and control voltage were swept; EMI was injected through different ports.

Variable	Value	Sweep?	Range Type	From	To	Step Mode	Total Steps	Inclusion List	Exclusion List
a	0.4	☒	From/To	0.2	0.6	Auto	3		
pho	150	☒	From/To	0	360	Auto	10		
v	0.9	☒	From/To	0	1.8	Auto	10		

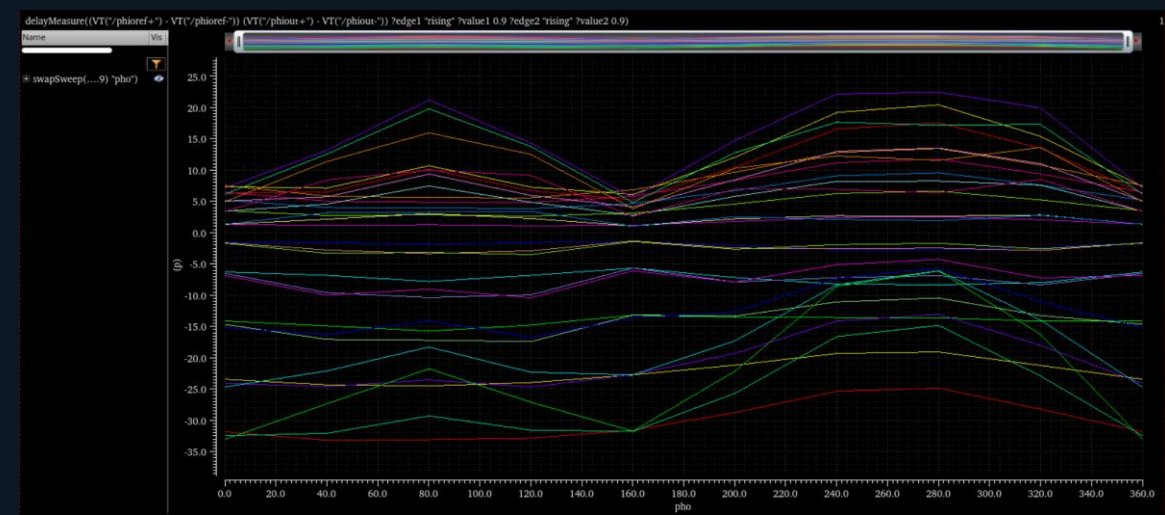
Sweep variables



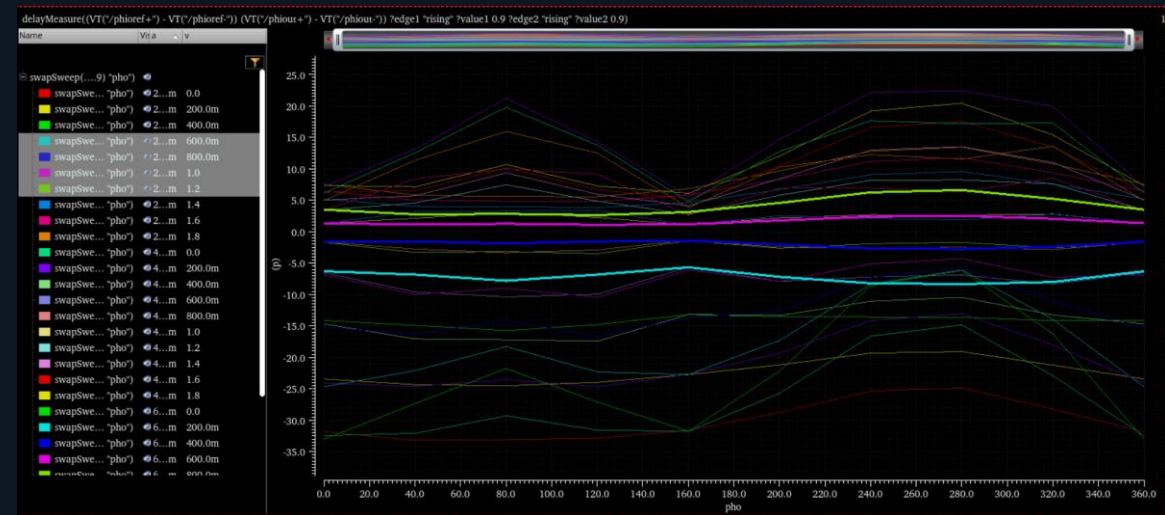
All-port / multi-port EMI testbench

Intermediate Architecture: Phase-Dependent Delay

The response remained irregular; this confirmed incomplete common-mode rejection.



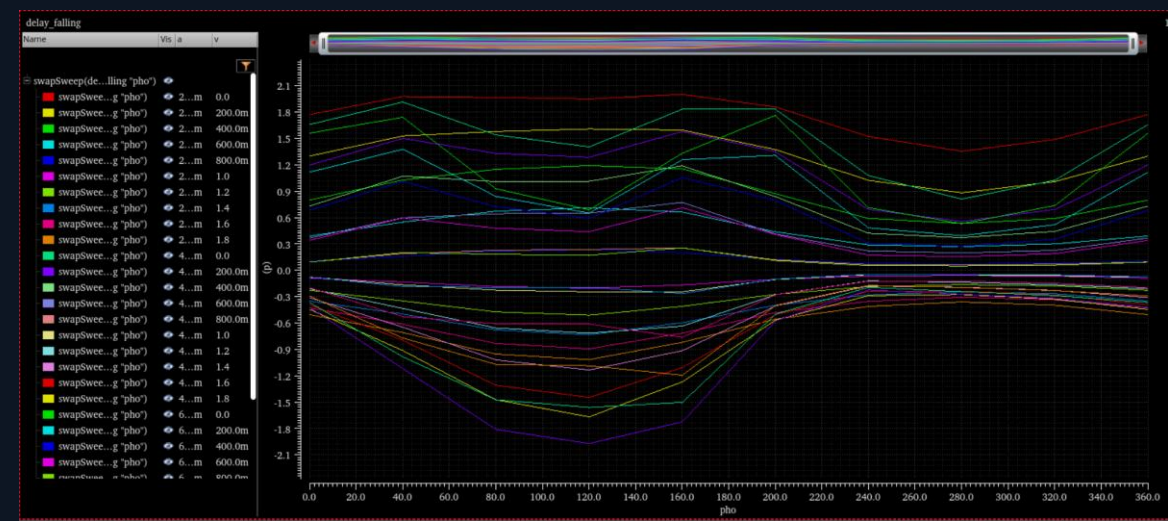
Delay vs phase



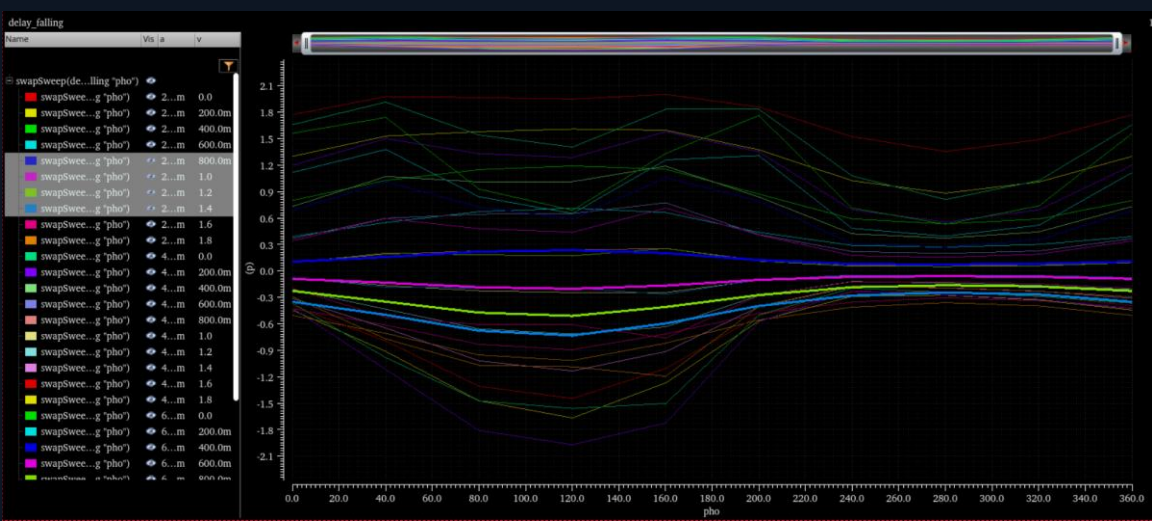
Expanded phase sweep

Intermediate Architecture: Falling-Edge Behaviour

Falling edge did not behave symmetrically with rising edge, making delay extraction unreliable.



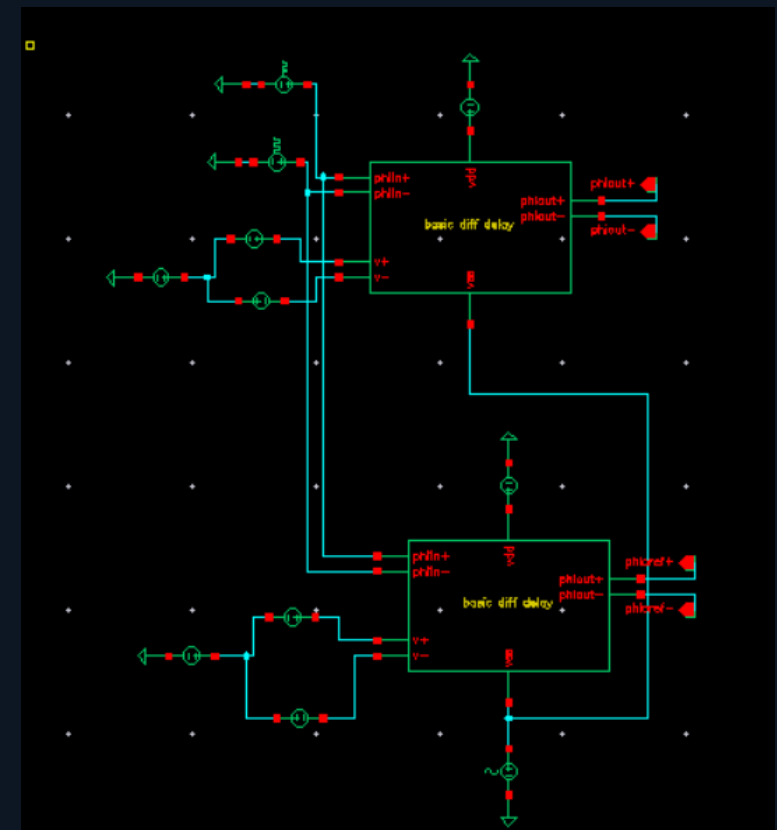
Falling-edge response



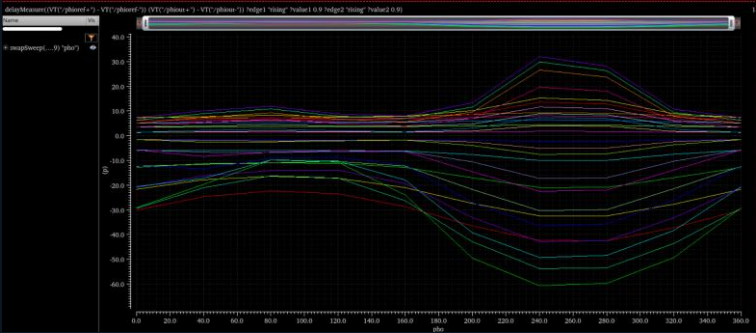
Additional falling-edge result

Supply and Combined EMI Paths

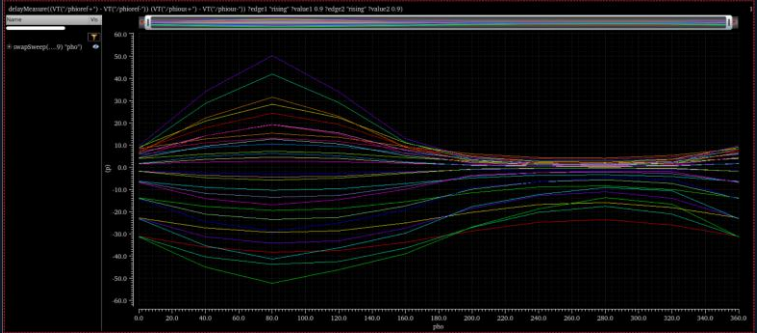
Supply EMI affected current and threshold; combined EMI produced residual jitter.



Supply EMI testbench



Supply EMI delay



All-port delay response

Intermediate Architecture: Mathematical Insight

$$\begin{aligned}\Delta t_d = & S_{in} \Delta V_{in} + S_{ref} \Delta V_{ref} \\ & + S_{DD} \Delta V_{DD} + S_{SS} \Delta V_{SS}\end{aligned}$$

If sensitivities are unequal: residual jitter remains

Design goal: reduce common-mode terms while preserving delay gain

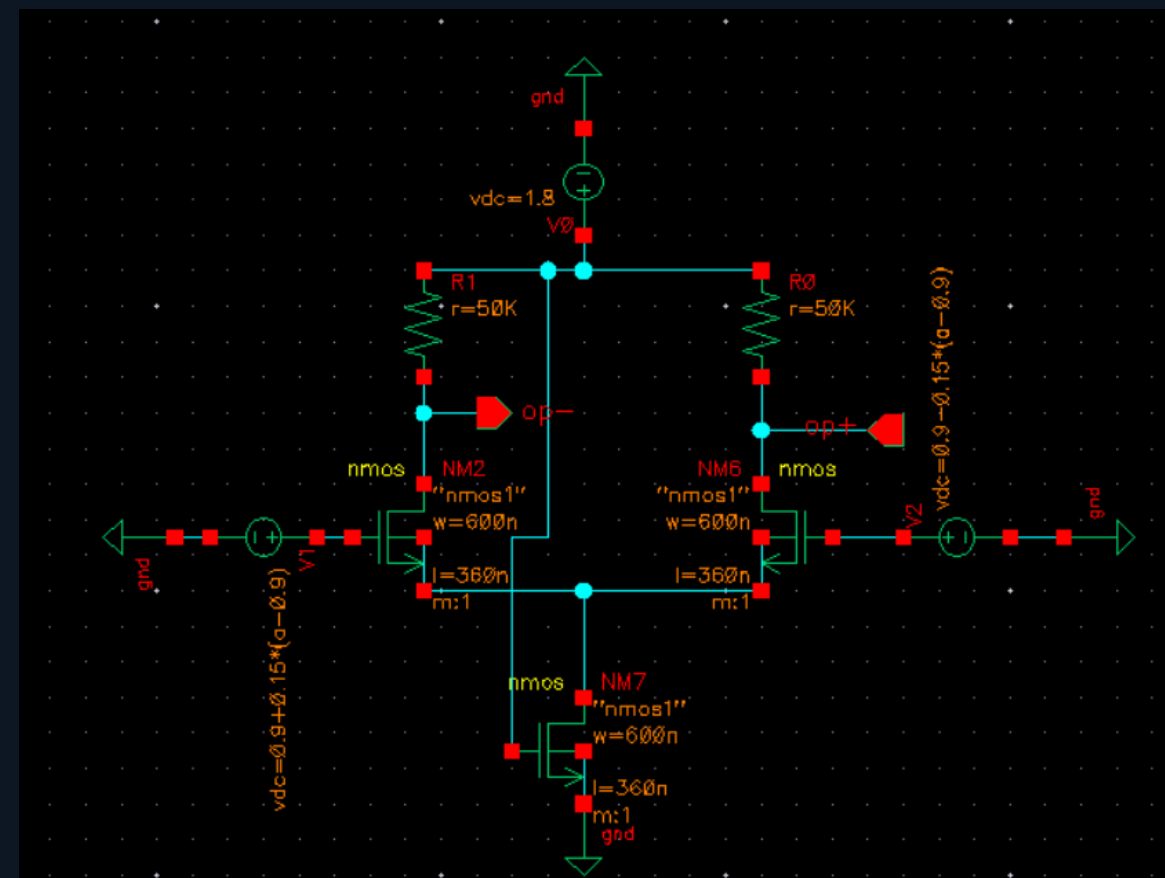
Delay error has multiple sensitivity paths, so only cancelling input common-mode noise is insufficient.

Final Design Direction

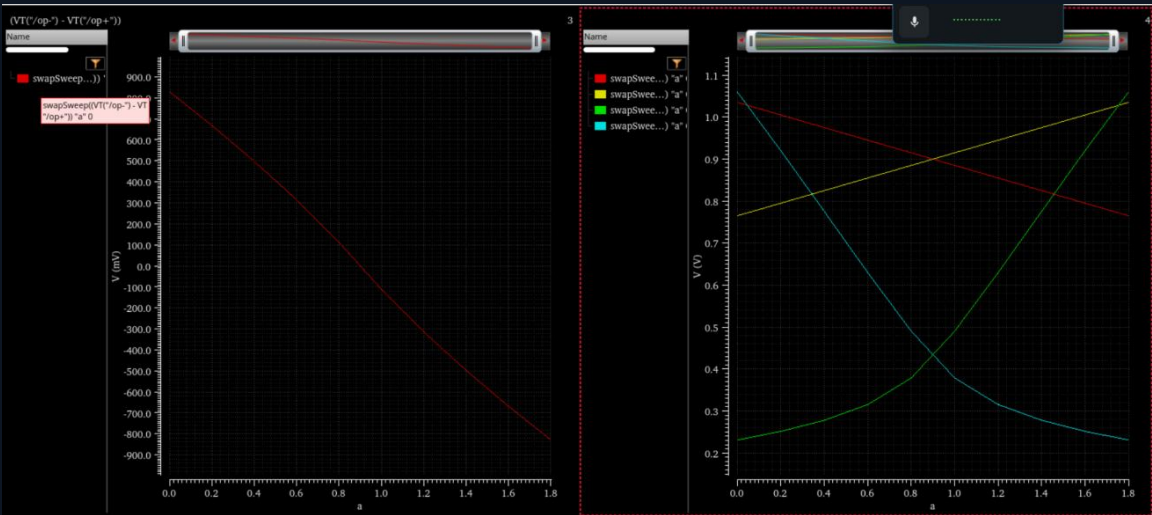
Compressed differential control around the valid operating point

Differential Pair DC Study

Direct 0-1.8 V control pushed the pair out of its valid small-signal/saturation region.



Differential pair schematic



DC output mapping

Compressed Control Mapping

$$V_{CM} = 0.9 \text{ V}$$

$$V_{plus} = 0.9 + k(A - 0.9)$$

$$V_{minus} = 0.9 - k(A - 0.9)$$

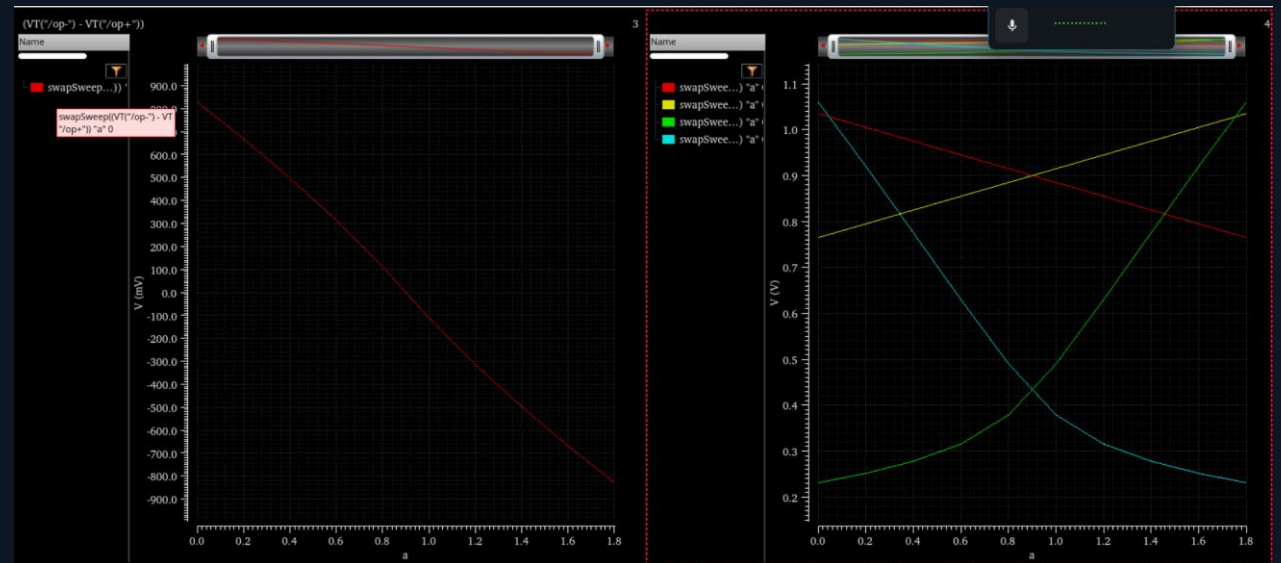
$$k \approx 0.15$$

$$v_{id} = 2k(A - 0.9)$$

Full rail-to-rail control A is mapped into a small differential swing around 0.9 V.

Why Compression Helped

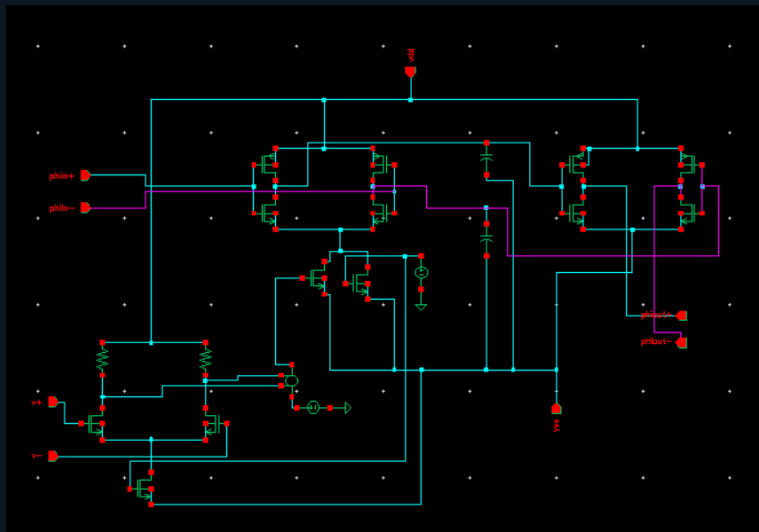
Both differential-pair transistors remain closer to saturation.
Common-mode cancellation assumption becomes more valid.
Low-amplitude EMI appears similarly on both branches.
Trade-off: compressed input reduces control sensitivity.



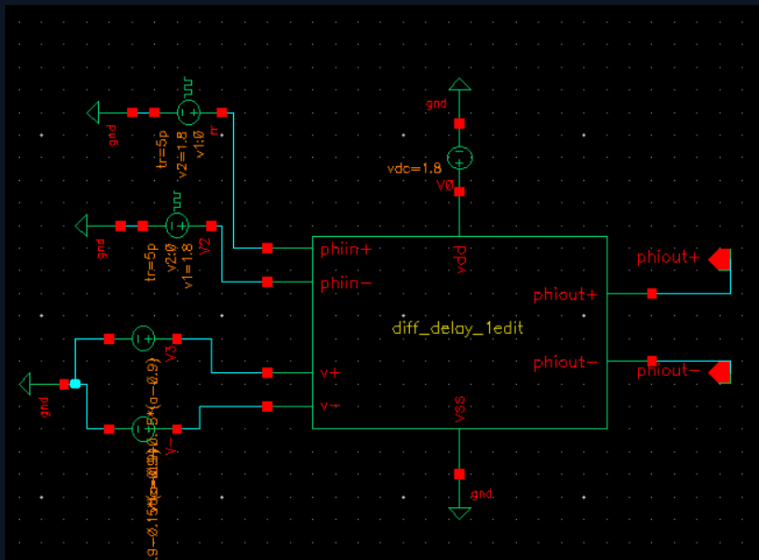
Compressed/small-signal operation around 0.9 V.

First Edit: Final Carried-Forward Cadence Direction

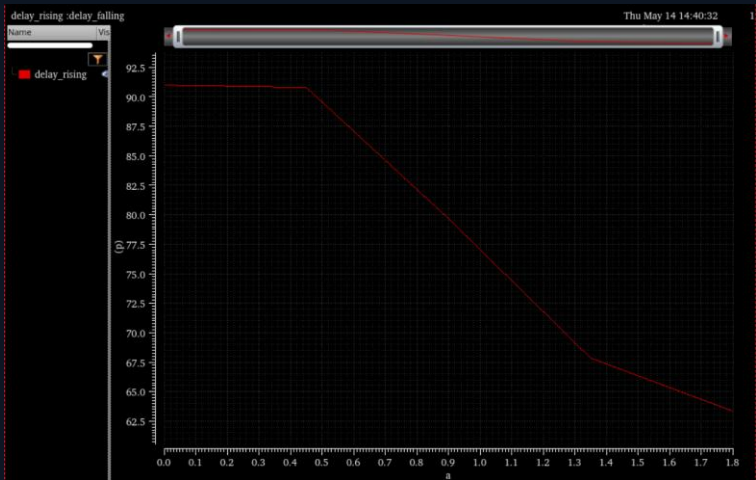
This is the actual Cadence-based architecture direction used in the final story. No AI-generated white diagram is used.



First edit schematic



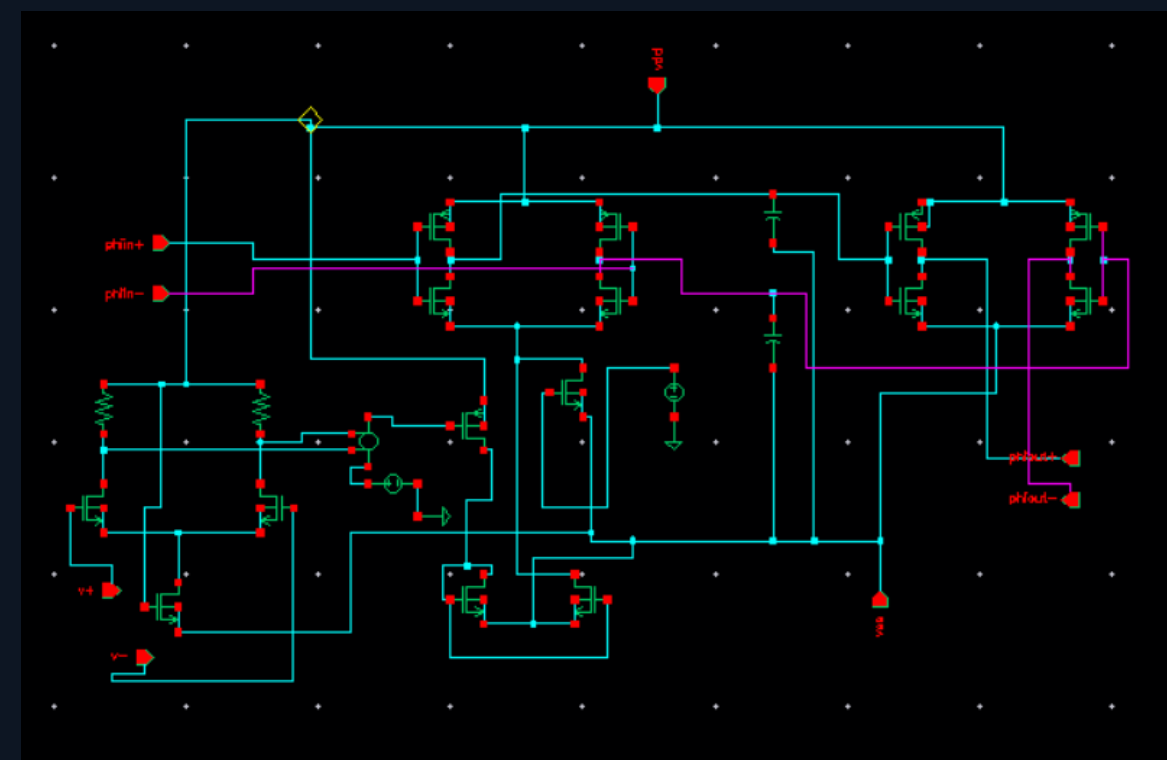
Delay response



EMI testbench/result

Second Edit: Ruled Out

Single-ended current mirror/tail-control attempt was rejected because delay became nearly constant.



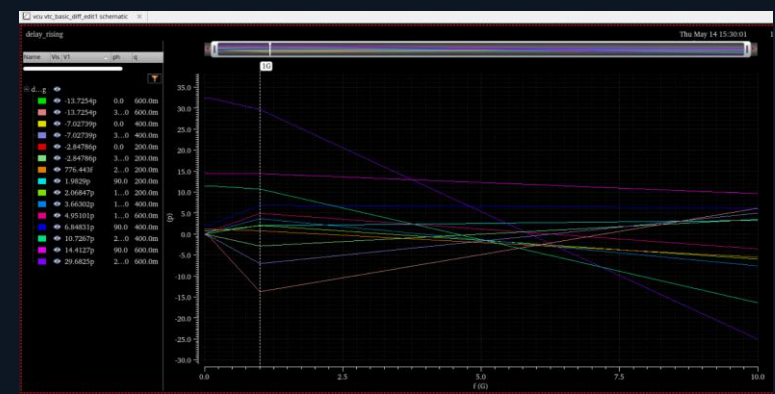
Second edit schematic



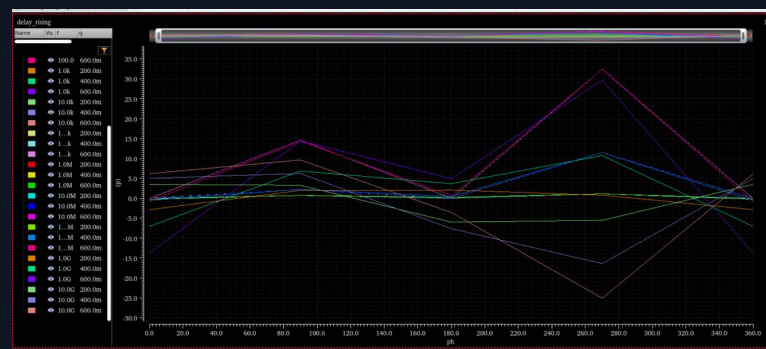
Nearly constant delay with sweep

Final EMI Observations

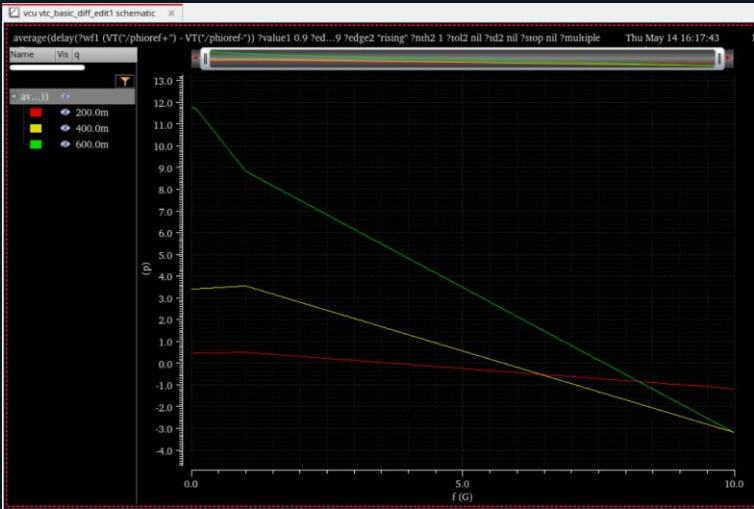
Low frequency / low amplitude: partial cancellation. High frequency / high amplitude: residual jitter remains.



Frequency and phase sweep



Frequency-dependent jitter



Average delay trend

Common-Mode Cancellation: Final Equation

$$V_{\text{plus}} = V_{\text{CM}} + v_{\text{id}}/2 + v_{\text{cm}}$$

$$V_{\text{minus}} = V_{\text{CM}} - v_{\text{id}}/2 + v_{\text{cm}}$$

$$V_{\text{plus}} - V_{\text{minus}} = v_{\text{id}}$$

$$T_{\text{plus}} = T(V_{\text{plus}}) + \text{delta_cm}$$

$$T_{\text{minus}} = T(V_{\text{minus}}) + \text{delta_cm}$$

$$\Delta T = T_{\text{plus}} - T_{\text{minus}}$$

Differential cancellation works only if both branches are matched and equally disturbed.

Small-Signal Delay Path

$$i_o = g_m v_d$$

$$v_b = A_v v_d$$

$$\Delta I = g_{m,\text{tail}} A_v v_d$$

$$t_d = C \Delta V / (I_0 + \Delta I)$$

This is the final analytical link from differential input to delay modulation.

$$\Delta t_d \approx -(t_{d0}/I_0) g_{m,\text{tail}} A_v v_d$$

Final BTP Contributions

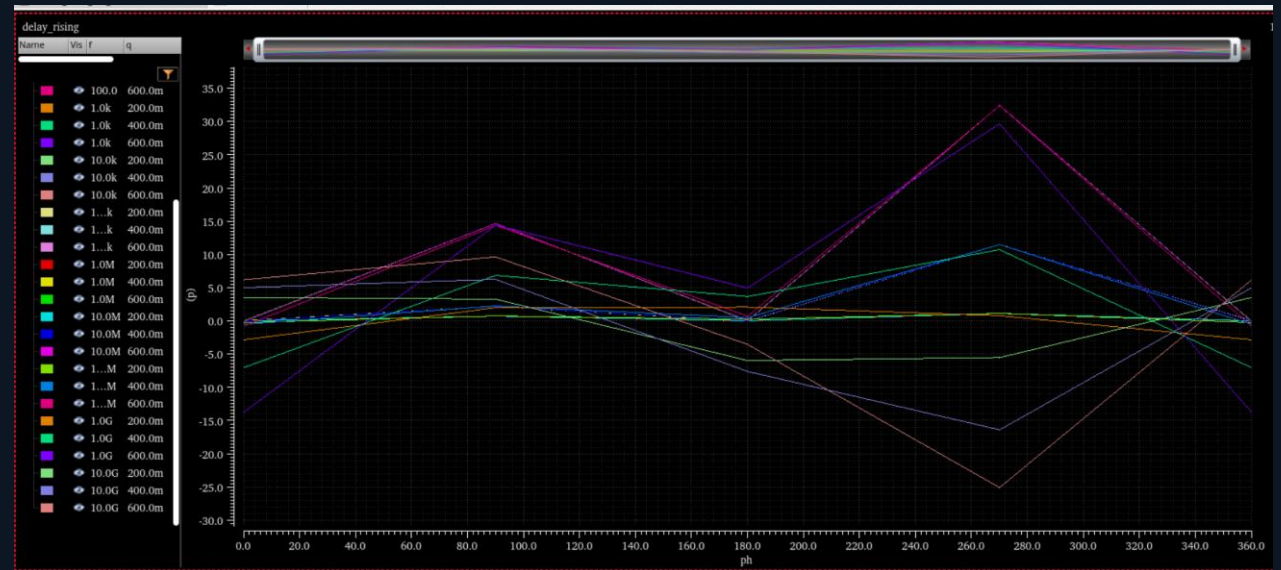
Documented pre-midsem baseline VTC, PVT and EMI behaviour.

Implemented and analyzed SET-H CDC and input-conditioned cells.

Identified why direct differential architectures failed.

Developed compressed differential control for partial EMI cancellation.

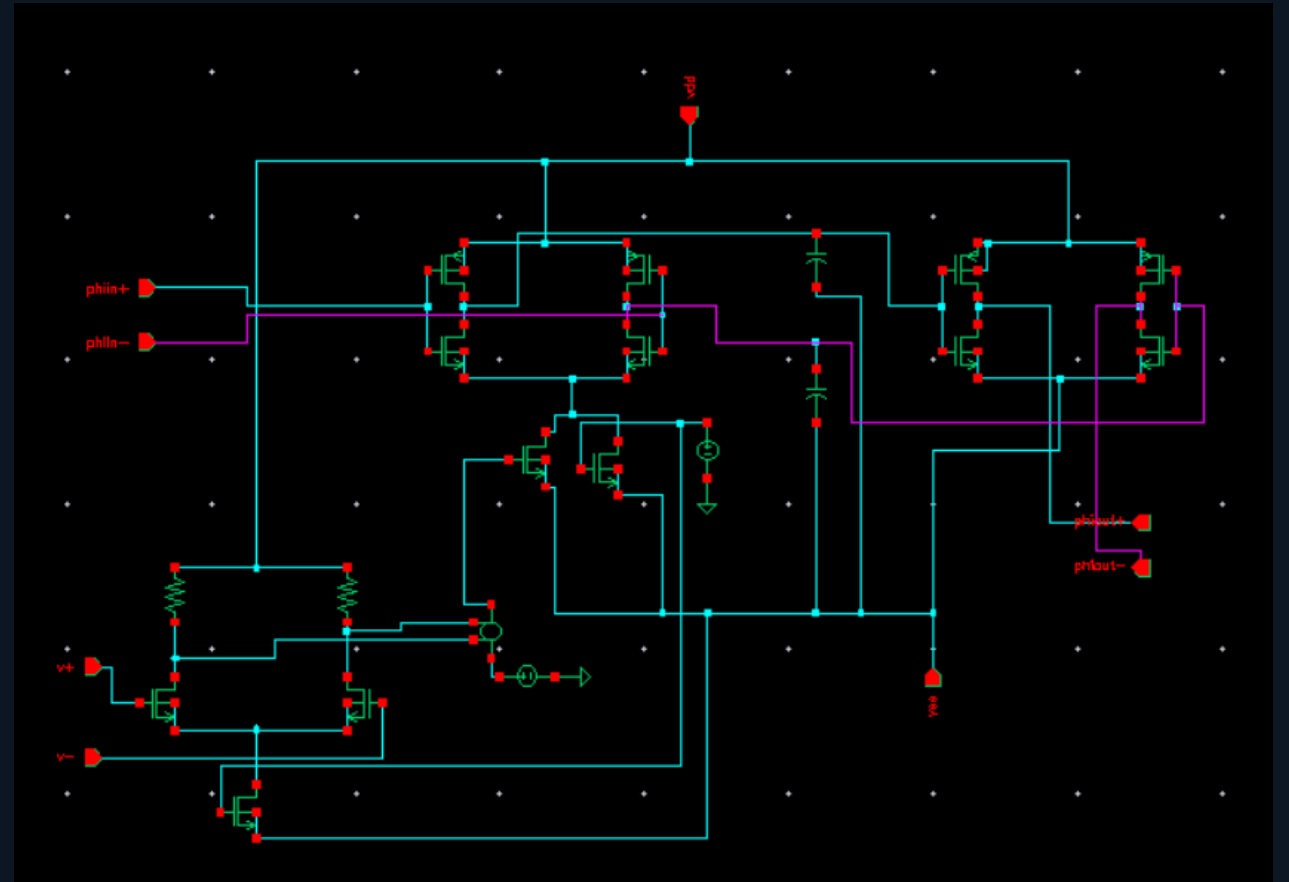
Explained remaining high-frequency limitations using CMRR and parasitics.



Final frequency-dependent residual jitter.

Final BTP Conclusion

EMI in a VTC becomes phase-dependent timing jitter.
Dual VCDU reduces common PVT variation, but not unequal EMI coupling.
Differential architecture helps only when biasing, matching and small-signal operation are preserved.
The final proposed direction is a Cadence-based compressed differential-control architecture.



Final carried-forward Cadence schematic.